

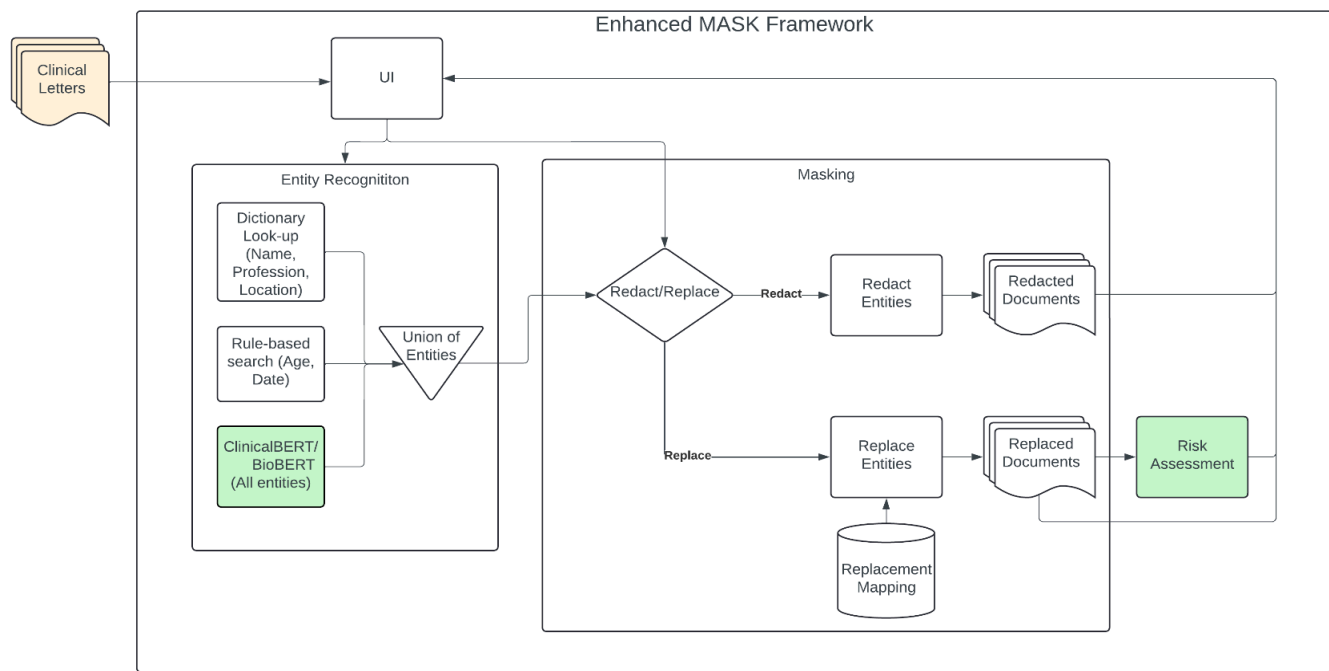
De-identifying Clinical Texts using Biomed-Clinical BERTs and Comprehensive Risk Assessment

13TH IEEE INTERNATIONAL CONFERENCE ON HEALTHCARE INFORMATICS

Motivations

- ▶ To accurately *identify* and *mask* sensitive entities (name, age, contact, IDs, location, etc.) on patients in clinical text while maintaining data utility for research.
- ▶ An advanced and user-friendly De-identification and risk assessment toolkit of clinical text by investigating state of the art language models from NLP.
- ▶ Investigating how biomedical and clinical domain-finetuned BERT models perform on this task in comparison to the traditional MASK framework.

DeID-Clinic Architecture: enhancing MASK with BioBERT and ClinicalBERT components



Methodology

- ▶ Transformer-based biomedical models **BioBERT** and **ClinicalBERT** vs other models including NER-BERT, NER-CRF, and LSTMs.
- ▶ Integrate BioBERT and ClinicalBERT into the **MASK** framework to facilitate effective de-identification (**MASK-BioBERT** / **ClinicalBERT**).
- ▶ Introduce a **risk assessment** tool that evaluates the potential re-identification risks of clinical data post de-identification.

Results: Different PLMs and F1-scores

Entity	BioBERT	ClinicalBERT	BiLSTM	CRF	Items
Name	98.5	97.0	97.0	96.0	622
Date	97.4	97.1	96.0	97.0	655
ID	98.8	98.9	92.0	97.0	75
Age	96.7	95.1	85.0	94.0	89
Location	92.8	92.1	91.0	90.0	278
Contact	94.1	95.4	94.0	96.0	69
Profession	74.8	73.0	79.0	63.0	27
Micro-Avg	96.5	95.9	95.0	95.0	1816
Macro-Avg	80.4	81.1	79.0	79.0	1816

Example Outputs and Interface:

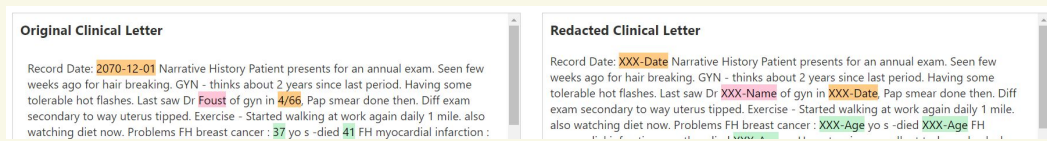
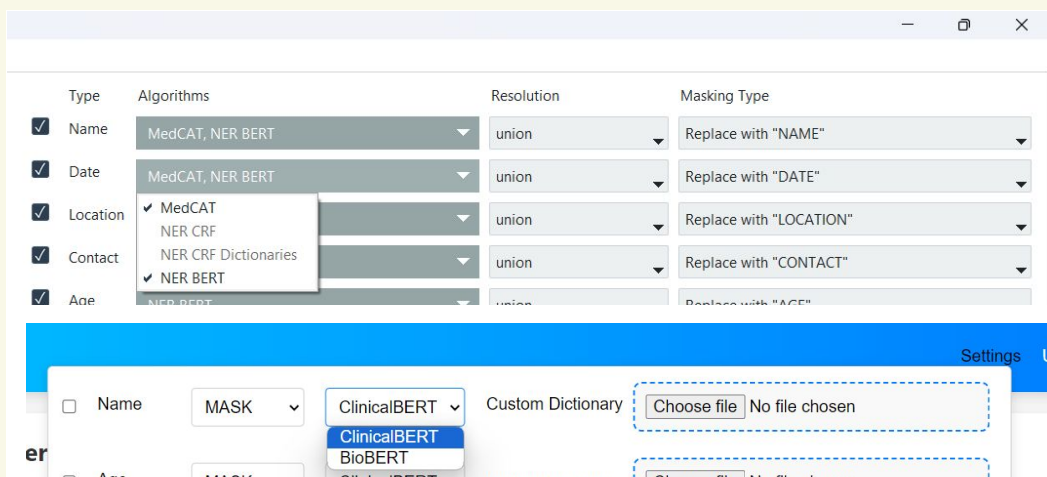


Figure: redacted clinical letter (up) and DeID-Clinic with MASK (below)



Outcomes:

- ▶ The BioBERT model produced highest Micro-average scores across all entity types, F1 96.5; the ClinicalBERT produced the highest Macro-average scores F1 81.1.
- ▶ The traditional models BiLSTM and CRF still won individual entity: **Profession** and **Contact**, respectively, F1 79.0 and 96.0, but lower on other entities. It is our future work plan, on how to improve model performances on these two entity types.
- ▶ A user-friendly risk-assessment platform is offered with UI (user-interface) design, which supports different type of files.
- ▶ Our fine-tuned model and source codes are openly available for public research purposes on Github.

Dhivin Shaji, Angel Paul, Lifeng Han*, Warren Del-Pinto, Goran Nenadic, Suzan Verberne

corresponding: l.han@lumc.nl Leiden Institute of Advanced Computer Science | <https://4dpicture.eu>

Leiden University Medical Centre | University of Manchester | model/code: <https://github.com/angelpaulm17/DeIDClinic>

